NLA Note

1 Background

Floating Point ($\mathbb{F}$): A number = mantissa $\times$ base$^{\text{exponent}}$ (e.g. $2048 = 1 \times 2^{11}$ mantissa=1, base=2, exponent=11)

Assumptions: There is $\varepsilon_m > 0$ **(machine epsilon)** $\varepsilon_m \approx 2.22 \times 10^{-16}$

A1: $\forall \alpha \in \mathbb{R}, \exists \varepsilon \in (-\varepsilon_m, \varepsilon_m)$ s.t. $fl: \mathbb{R} \to \mathbb{F}$ by $fl(\alpha) = \alpha(1+\varepsilon)$

A2: $\forall \alpha, \beta \in \mathbb{F}, \forall * \in \{+,-,\times,\div\}, \exists \varepsilon \in (-\varepsilon_m, \varepsilon_m)$ s.t. $\alpha \circledast \beta = (\alpha * \beta)(1+\varepsilon)$

A1: The rounding error in α is relative to the size of α ($|\alpha - fl(\alpha)| < |\alpha|\varepsilon_m$) A2: The error in $\alpha \circledast \beta$ is relative to the size of $\alpha * \beta$ ($|\alpha \circledast \beta - \alpha * \beta| \le |\alpha * \beta|\varepsilon_m$)

Cost of Algorithm (C): $C :=$ number of floating point operations (FLOPs) If $C(n) \le \alpha f(n), \Rightarrow C(n) = \mathcal{O}(f(n))$.

We consider that: 交换/赋值/相反数 no cost (i.e. Permutation $\mathbf{P}$ is free), 比较两数 1 FLOPs, $\sqrt{\ }$ 1 FLOPs.

Norms: $\|\mathbf{x}\|_p := (\sum_{i=1}^n |x_i|^p)^{1/p}, p \ge 1$ $\|\mathbf{x}\|_\infty := \max_{1\le i\le n} |x_i|$ $\|\mathbf{x}\|_2 := (\sum_{i=1}^n |x_i|^2)^{1/2}$ $\|\mathbf{x}\|_1 := \sum_{i=1}^n |x_i|$

1. $\|\mathbf{x}\|_p \ge 0$ $\|\mathbf{x}\|_p = 0 \Leftrightarrow \mathbf{x} = \mathbf{0}$
2. $\|\alpha\mathbf{x}\|_p = |\alpha|\|\mathbf{x}\|_p$ for $\alpha \in \mathbb{R}$
3. $\|\mathbf{x}+\mathbf{y}\|_p \le \|\mathbf{x}\|_p + \|\mathbf{y}\|_p$ (Triangle inequality)
4. **Induced Norm:** $\|\mathbf{A}\|_p := \sup_{\mathbf{x}\ne 0} \frac{\|\mathbf{A}\mathbf{x}\|_p}{\|\mathbf{x}\|_p} = \sup_{\|\mathbf{x}\|_p=1} \|\mathbf{A}\mathbf{x}\|_p$

1. $\|\mathbf{A}_{m\times n}\|_\infty := \max_{1\le j\le m} \sum_{i=1}^n |a_{ij}|$ (maximum row sum)
2. $\|\mathbf{A}_{m\times n}\|_1 := \max_{1\le i\le n} \sum_{j=1}^m |a_{ij}|$ (maximum column sum)
3. $\|\mathbf{A}_{m\times n}\|_2 := \sqrt{\rho(\mathbf{A}^\top\mathbf{A})}$ $\rho(\mathbf{A}^\top\mathbf{A}) = \max\{|\lambda| : \lambda \text{ is eigenvalue of } \mathbf{A}^\top\mathbf{A}\}$

For Matrix: $\|\mathbf{A}\mathbf{x}\|_p \le \|\mathbf{A}\|_p\|\mathbf{x}\|_p$ $\|\mathbf{A}\mathbf{B}\|_p \le \|\mathbf{A}\|_p\|\mathbf{B}\|_p$ $\|\mathbf{A}^k\|_p \le \|\mathbf{A}\|_p^k$ $\|\mathbf{A}+\mathbf{B}\|_p \le \|\mathbf{A}\|_p + \|\mathbf{B}\|_p$

2 Methods for Systems of Linear Equations (SLE)

Ideal: We want to solve SLE: $\mathbf{A}\mathbf{x} = \mathbf{b}$, where $\mathbf{A}_{n\times n}$ is invertible, $\mathbf{x}, \mathbf{b} \in \mathbb{R}^n$

2.1 Backward Stable | Condition Number | Error

Backward Stable: Algorithm solve SLE is Backward Stable if: $\exists \alpha(n) > 0$ s.t. $\forall \mathbf{A}, \mathbf{b}$, computed solution $\hat{\mathbf{x}}$ has: $(\mathbf{A}+\Delta\mathbf{A})\hat{\mathbf{x}} = \mathbf{b} + \Delta\mathbf{b}$ and $\frac{\|\Delta\mathbf{A}\|_p}{\|\mathbf{A}\|_p} + \frac{\|\Delta\mathbf{b}\|_p}{\|\mathbf{b}\|_p} \le \alpha\varepsilon_m$ ε_m : machine epsilon.

Condition Number ($\kappa_p(\mathbf{A})$): $\kappa_p(\mathbf{A}) := \begin{cases} \|\mathbf{A}\|_p\|\mathbf{A}^{-1}\|_p \ge 1 & \text{if } \mathbf{A} \text{ is invertible} \\ \infty & \text{otherwise} \end{cases}$ $\begin{cases} \text{If } \kappa_p(\mathbf{A}) \text{ is large (e.g. } \kappa_p(\mathbf{A}) \ge 10^{12} \approx \varepsilon_m^{-1}), & \mathbf{A} \text{ is ill-conditioned.} \\ \text{If } \kappa_p(\mathbf{A}) \le 10^4, & \mathbf{A} \text{ is well-conditioned.} \end{cases}$ ps: $\kappa_p(\mathbf{A})$ 衡量矩阵奇异性

If $\mathbf{A}_{n\times n}$ s.t. $|\lambda_1| \ge |\lambda_2| \ge \cdots \ge |\lambda_n|$ corresponding $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n$ are orthonormal set. **(e.g. $A^\top = A$)** Then $\kappa_2(\mathbf{A}) = \|\mathbf{A}\|_2\|\mathbf{A}^{-1}\|_2 = \frac{|\lambda_1|}{|\lambda_n|} \ge 1$

Relative Error: $\frac{\|\hat{\mathbf{x}}-\mathbf{x}\|_p}{\|\mathbf{x}\|_p}$ **Actual Error:** $\|\hat{\mathbf{x}}-\mathbf{x}\|_p$ Let $\mathbf{A}\mathbf{x} = \mathbf{b}$ and $(\mathbf{A}+\Delta\mathbf{A})\hat{\mathbf{x}} = \mathbf{b}$. If $\kappa_p(\mathbf{A})\frac{\|\Delta\mathbf{A}\|_p}{\|\mathbf{A}\|_p} < 1$, then: $\frac{\|\mathbf{x}-\hat{\mathbf{x}}\|_p}{\|\mathbf{x}\|_p} \le \frac{\kappa_p(\mathbf{A})}{1-\kappa_p(\mathbf{A})\frac{\|\Delta\mathbf{A}\|_p}{\|\mathbf{A}\|_p}} \cdot \frac{\|\Delta\mathbf{A}\|_p}{\|\mathbf{A}\|_p}$

Generally, Let $\mathbf{A}\mathbf{x} = \mathbf{b}$ and $(\mathbf{A}+\Delta\mathbf{A})\hat{\mathbf{x}} = \mathbf{b}+\Delta\mathbf{b}$. If $\kappa_p(\mathbf{A})\frac{\|\Delta\mathbf{A}\|_p}{\|\mathbf{A}\|_p} < 1$ then $\frac{\|\mathbf{x}-\hat{\mathbf{x}}\|_p}{\|\mathbf{x}\|_p} \le \frac{\kappa_p(\mathbf{A})}{1-\kappa_p(\mathbf{A})\frac{\|\Delta\mathbf{A}\|_p}{\|\mathbf{A}\|_p}} \cdot \left(\frac{\|\Delta\mathbf{A}\|_p}{\|\mathbf{A}\|_p} + \frac{\|\Delta\mathbf{b}\|_p}{\|\mathbf{b}\|_p}\right)$ ps: $\frac{\|\Delta\mathbf{A}\|_p}{\|\mathbf{A}\|_p}$ $\frac{\|\Delta\mathbf{b}\|_p}{\|\mathbf{b}\|_p}$ 小代表 good stability.

2.2 LU Factorization | Gaussian Elimination (GE)

Def of LU: If $\mathbf{A}_{n\times n}$ is invertible, then $^1\exists\,\mathbf{L}$: 单位下三角, $^2\exists\,\mathbf{U}$: 可逆上三角. s.t. $\mathbf{A} = \mathbf{L}\mathbf{U}$ **(unique)**

LU Factorization: By Gaussian elimination, $\mathbf{A}_{n\times n} = (\prod_{k=1}^{n-1} \mathbf{L}_k)^{-1}\mathbf{U} =: \mathbf{L}\mathbf{U}$ $\mathbf{L}_k$: Lower triangular with 1s on the diagonal (by row operation on single column)

1. 如果 $\mathbf{L}$ 是单位下三角矩阵, 且只有第 k 列有非零元素在对角线下, 则 $\mathbf{L}^{-1}$ 矩阵是 $\mathbf{L}$ 对角线不动, 其余为相反数.
2. 如果 $\mathbf{L}_1, \mathbf{L}_2$ 矩阵是单位下三角矩阵, 且 $\mathbf{L}_1|\mathbf{L}_2$ 仅在第 $1 \sim k|k+1 \sim n$ 列有非零元素在对角线下, 则 $\mathbf{L}_1\mathbf{L}_2$ 矩阵是他们的" 拼接".

Error Analysis of GE: Gaussian elimination is an **unstable** algorithm. (i.e. floating point arithmetic can have a disastrous effect on the computed solution.)

2.3 LUPP | LUCP | GEPP | GECP

Permutation Matrix: For a permutation π, $\mathbf{P}_\pi := [\mathbf{e}_{\pi(1)}, \ldots, \mathbf{e}_{\pi(n)}]$ ps: 写的时候是对 I, 第 i 行的 1 移动到第 $\pi(i)$ 行. 注意: 用另一方式定义的 $\mathbf{P}_\pi$ 会让后面对 π 结论不一样

· 左边乘 $\mathbf{P}_\pi$: 交换行 ($\mathbf{P}_\pi$ 的行 代表具体行的交换方法, π 也代表交换行的方法) · 右边乘 $\mathbf{P}_\pi$: 交换列 ($\mathbf{P}_\pi$ 的列 代表具体列的交换方法, π^{-1} 才代表交换列的方法)

· Proposition: $\mathbf{P}_\pi$ is orthogonal, i.e. $\mathbf{P}_\pi^\top = \mathbf{P}_\pi^{-1}$ · 对于二阶置换矩阵 **(Involution)**: $\mathbf{P}_\pi^2 = \mathbf{I}$, i.e. $\mathbf{P}_\pi^{-1} = \mathbf{P}_\pi$ · **GEPP|GECP:** 交换行 (行和列) 使每次 $|u_{kk}|$ 最大**(stable)**

Backward Stable of GEPP: $\forall$ SLE sol $\hat{\mathbf{x}}$ by GEPP, $\exists \Delta\mathbf{A}$ s.t. $(\mathbf{A}+\Delta\mathbf{A})\hat{\mathbf{x}} = \mathbf{b}$ and $\frac{\|\Delta\mathbf{A}\|_\infty}{\|\mathbf{A}\|_\infty} \le p(n)2^{n-1}\varepsilon_m$ $p(n)$ is a cubic polynomial

2.4 QR Factorization Method

QR Factorization: QR factorization of $\mathbf{A}_{n\times n}$ is $\mathbf{A} = \mathbf{Q}\mathbf{R}$, where $\mathbf{Q}$ is orthogonal (i.e. $\mathbf{Q}^T = \mathbf{Q}^{-1}$) and $\mathbf{R}$ is non-singular upper triangular.

· Def of $\mathbf{Q}$ is orthogonal is equivalent to $\mathbf{Q}^T\mathbf{Q} = \mathbf{I}$. (i.e. columns of $\mathbf{Q}$ forming an orthonormal set.) · QR produce **large errors**

· **Orthogonal Projection:** $\text{proj}_{\mathbf{u}}(\mathbf{x}) = \frac{\mathbf{x}\cdot\mathbf{u}}{\|\mathbf{u}\|_2^2}\mathbf{u} = (\mathbf{u}^\top\mathbf{x})\mathbf{u}$ if $\|\mathbf{u}\|_2 = 1$. $\mathbf{x} = \text{proj}_{\mathbf{u}}(\mathbf{x})^{\text{parallel to } \mathbf{u}} + (\mathbf{x} - \text{proj}_{\mathbf{u}}(\mathbf{x}))^{\text{orthogonal to } \mathbf{u}}$

Error Analysis of MGS: The matrices $\widehat{\mathbf{Q}}$ and $\widehat{\mathbf{Q}}$ computed by MGS satisfies: $\widehat{\mathbf{Q}}^\top\widehat{\mathbf{Q}} = \mathbf{I} + \Delta\mathbf{I}$ $\widehat{\mathbf{Q}}\widehat{\mathbf{R}} = \mathbf{A} + \Delta\mathbf{A}$ where:

$\|\Delta\mathbf{I}\|_2 \le \frac{\alpha_1(n)\varepsilon_m\kappa_2(\mathbf{A})}{1-\alpha_1(n)\varepsilon_m\kappa_2(\mathbf{A})}$, if $1-\alpha_1(n)\varepsilon_m\kappa_2(\mathbf{A}) > 0$ $\|\Delta\mathbf{A}\|_2 \le \alpha_2(n)\varepsilon_m\|\mathbf{A}\|_2$ ps: $\alpha_1(n), \alpha_2(n)$ 只由 n 决定

2.5 Householder QR factorisation

Householder QR: (Most) **stable** algorithm **Householder Reflection Matrix:** $\mathbf{H}(\mathbf{w}) := \mathbf{I} - 2\frac{\mathbf{w}\mathbf{w}^\top}{\|\mathbf{w}\|_2^2}$ $\mathbf{H}(\mathbf{w})\mathbf{x} = \mathbf{x} - 2\text{proj}_{\mathbf{w}}(\mathbf{x})$

· $\mathbf{H}(\mathbf{w})$ is **orthogonal** and **symmetric** (In the Algorithm is $\mathbf{Q}_k, \mathbf{H}_k$) · $\mathbf{H}(\mathbf{w})\mathbf{x}$ 作用为让 $\mathbf{x}$ 向量对 $\mathbf{w}$ 的法向量 (超平面) 上进行反射

Error Analysis of Householder QR: Let $\hat{\mathbf{x}}$ denote sol of $\mathbf{A}\mathbf{x} = \mathbf{b}$ by Householder QR+SQR. Then $(\mathbf{A}+\Delta\mathbf{A}) = \mathbf{b}$, where $\|\Delta\mathbf{A}\|_2 \le \alpha n^{\frac{5}{2}}\varepsilon_m\|\mathbf{A}\|_2$ α small constant

2.6 Iterative Methods for SLE

Iterative Method: For SLE $\mathbf{A}\mathbf{x} = \mathbf{b}$, if $\mathbf{A} = \mathbf{M} + \mathbf{N}$ with $\mathbf{M}$ is *simpler structure* than $\mathbf{A}$. (例如: 对角/三角/对称矩阵) $\Rightarrow$ 迭代求解近似解 Diff choices $\mathbf{M}, \mathbf{N} \to$ different iterative methods.

Error Analysis: Error $\mathbf{e}_k := \mathbf{x} - \mathbf{x}_k \Rightarrow \mathbf{e}_k = \mathbf{R}^k\mathbf{e}_0$ $\mathbf{R} := -\mathbf{M}^{-1}\mathbf{N}$ **Theorem:** If $\|\mathbf{R}\|_p < 1$, then $\lim_{k\to\infty} \|\mathbf{e}_k\|_p = 0 \Leftrightarrow \lim_{k\to\infty} \mathbf{x}_k = \mathbf{x}$ (Convergence) **GAU 收敛快于 JAC**

JAC|GAU: If $\mathbf{A}$ is **strictly diagonally dominant** [i.e. $|a_{ii}| > \sum_{j\ne i}|a_{ij}|\ \forall i$ (对角线元素大于该行其他元素之和)], then converges for $p = \infty$. For $p = 1$ 看列和/转置.

3 Polynomial Fitting & LSQ

ps: 范德蒙德矩阵 ill-cond

Vandermonde Matrix: $A \in \mathbb{R}^{n\times m}$ s.t. $a_{ij} = a_i^{j-1}$ **Poly:** Deg $m-1$ poly fit to points: $\{(a_i, b_i)\}_{i=1}^n$ by solving $\mathbf{A}\mathbf{x} = \mathbf{b} \Rightarrow$ sol $\mathbf{x}$ is coeff. of poly $p(x) = \sum_{j=1}^m x_j x^{j-1}$

LSQ: Finding *line* which best fits $\{(a_i, b_i)\}_{i=1}^n$ by *minimise* $\|\mathbf{A}\mathbf{x}-\mathbf{b}\|_2$, $A \in \mathbb{R}^{n\times 2}$ vandermonde. **Thm:** *minimises* $\|\mathbf{A}\mathbf{x}-\mathbf{b}\|_2 \Leftrightarrow$ solve **Normal Equations** $\mathbf{A}^\top\mathbf{A}\mathbf{x} = \mathbf{A}^\top\mathbf{b}$

Moore-Penrose Pseudo-inverse of A: $\mathbf{A}^\dagger = (\mathbf{A}^\top\mathbf{A})^{-1}\mathbf{A}^\top$ when $\text{rank}(\mathbf{A}) = m$, $(\mathbf{A}^\top\mathbf{A})^{-1}$ exist · *Normal Equations* has **unique sol** if $\mathbf{A}^\top\mathbf{A}$ invertible. · If $\mathbf{A}$ invertible, $\mathbf{A}^\dagger = \mathbf{A}^{-1}$

3.1 Singular Value Decomposition (SVD)

New Def of Condition Number: If $\mathbf{A}_{m\times n}, m \ge n$ s.t. it's full rank (i.e. $\text{rank}(\mathbf{A}) = n$), then $\kappa_p(\mathbf{A}) := \|\mathbf{A}\|_p\|\mathbf{A}^\dagger\|_p \ge 1$ $\kappa_2(\mathbf{A}^\top\mathbf{A}) = \kappa_2(\mathbf{A})^2$ $\kappa_2(\mathbf{A}) = \frac{\sigma_{\max}}{\sigma_{\min}}$

SVD: $\mathbf{A}_{m\times n} = \mathbf{U}\Sigma\mathbf{V}^\top$ where $\mathbf{U}_{m\times m}, \mathbf{V}_{n\times n}$ *orthogonal*, $\Sigma_{m\times n}$ diagonal with singular values $\sigma_1, \ldots, \sigma_p$, $p = \min\{m, n\}$ $\mathbf{U}/\mathbf{V}$ 的列为 left/right singular vectors.

^{1} Always exist ^{2} $\mathbf{A}\mathbf{v}_i = \sigma_i\mathbf{u}_i$ and $\mathbf{A}^\top\mathbf{u}_i = \sigma_i\mathbf{v}_i$ ^{3} $\sigma_i^2|\mathbf{v}_i$ **are eigenvalues/vectors of $\mathbf{A}^\top\mathbf{A}$** (i.e. $\mathbf{A}^\top\mathbf{A}\mathbf{v}_i = \sigma_i^2\mathbf{v}_i$) ^{4} $\text{rank}(\mathbf{A}) = \text{rank}(\Sigma) = \#$ non-zero σ_i ^{5} $\mathbf{u}_i = \frac{1}{\sigma_i}\mathbf{A}\mathbf{v}_i$

Reduced QR Factorisation: By QR: $\mathbf{A}_{m\times n} = \mathbf{Q}_{m\times m}\mathbf{R}_{m\times n}, m \ge n$. ^{1} R 的 n 行以下全 0 ^{2} $\mathbf{Q} = [\mathbf{Q}_1\ \mathbf{Q}_2]$, $\mathbf{R} = [\mathbf{R}_1\ \mathbf{0}]^\top \Rightarrow \mathbf{A} = \mathbf{Q}_1\mathbf{R}_1$ where $\mathbf{Q}_1, \mathbf{R}_1 \in \mathbb{R}^{n\times n}$

Reduced SVD: By SVD: $\mathbf{A} = \mathbf{U}\Sigma\mathbf{V}^\top, m \ge n$. ^{1} If $\mathbf{B}$ orthogonal, then $\mathbf{B}^{-1} = \mathbf{B}^\top$, $\det(\mathbf{B}) = \pm 1$ and $\|\mathbf{B}\mathbf{x}\|_2 = \|\mathbf{x}\|_2, \forall \mathbf{x}$ ^{2} $\mathbf{U} = [\mathbf{U}_1\ \mathbf{U}_2]$, $\Sigma = [\Sigma_1\ \mathbf{0}]^\top \Rightarrow \mathbf{A} = \mathbf{U}_1\Sigma_1\mathbf{V}^\top$, $\mathbf{U}_1 \in \mathbb{R}^{m\times n}, \Sigma_1 \in \mathbb{R}^{n\times n}$

Error Analysis of LSQ-QR: Let $\hat{\mathbf{x}}$ denote sol of LSQ-QR (Householder). Then $\hat{\mathbf{x}}$ minimises $\|(\mathbf{A}+\Delta\mathbf{A})\mathbf{x} - \mathbf{b}\|_2$, and $\frac{\|\Delta\mathbf{A}\|_2}{\|\mathbf{A}\|_2} \le \alpha n^{\frac{3}{2}} m\varepsilon_m, \alpha > 0$

Error Analysis LSQ (any): Let $\mathbf{x}|\hat{\mathbf{x}}$ minimises $\|\mathbf{A}\mathbf{x}-\mathbf{b}\|_2 | \|(\mathbf{A}+\Delta\mathbf{A})\mathbf{x}-\mathbf{b}\|_2$ and $\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{A}+\Delta\mathbf{A}) = n$. If $\kappa_2(\mathbf{A})\frac{\|\Delta\mathbf{A}\|_2}{\|\mathbf{A}\|_2} < 1 \Rightarrow \frac{\|\mathbf{x}-\hat{\mathbf{x}}\|_2}{\|\mathbf{x}\|_2} \le \frac{\kappa_2(\mathbf{A})}{1-\kappa_2(\mathbf{A})\frac{\|\Delta\mathbf{A}\|_2}{\|\mathbf{A}\|_2}} \cdot \frac{\|\Delta\mathbf{A}\|_2}{\|\mathbf{A}\|_2}\left(2 + (\kappa_2(\mathbf{A})+1)\frac{\|\mathbf{r}\|_2}{\|\mathbf{A}\|_2\|\mathbf{x}\|_2}\right)$, $\mathbf{r} = \mathbf{b} - \mathbf{A}\mathbf{x}$

4 Eigenvalue Problems

(In this section: $|\lambda_1| \ge |\lambda_2| \ge \cdots \ge |\lambda_n|$ are eigenvalues of $\mathbf{A} \in \mathbb{R}^{n\times n}$ with corresponding eigenvectors $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n$. $\lambda_1 = \lambda_{\max} \leftrightarrow \mathbf{x}_{\max}$, $\lambda_n = \lambda_{\min} \leftrightarrow \mathbf{x}_{\min}$)

Rayleigh Quotient: $r_{\mathbf{A}}(\mathbf{x}) = \frac{\mathbf{x}^\top\mathbf{A}\mathbf{x}}{\mathbf{x}^\top\mathbf{x}}$ $r_{\mathbf{A}}(\mathbf{x}_i) = \lambda_i$ Eigenvalues $\leftrightarrow$ Eigenvectors: $\mathbf{x}_i \to \lambda_i$: $r_{\mathbf{A}}(\mathbf{x}_i) = \lambda_i$ $\lambda_i \to \mathbf{x}_i$: $(\mathbf{A} - \lambda_i\mathbf{I})\mathbf{x}_i = \mathbf{0}$ **Basis:** $\mathbf{A}\mathbf{x} = \sum_i \alpha_i\lambda_i\mathbf{x}_i$

Convergence of Power Iteration: If $|\lambda_1| > |\lambda_2|$ and $\mathbf{x}_1 \cdot \mathbf{z}^{(0)} \ne 0$, then $\exists {\sigma_k}_{k\in\mathbb{N}}$ with $\sigma_k = \pm 1$ s.t. $\lim_{k\to\infty} \|\sigma_k\mathbf{z}^{(k)} - \mathbf{x}_1\|_2 = 0$ and $\lim_{k\to\infty} |\lambda^{(k)} - \lambda_1| = 0$

Useful Tool: Eigenvalues of $\mathbf{A}^{-1}$ are $\mu_i = \frac{1}{\lambda_i}$ and $\lambda_{\max} = \mu_{\min}^{-1}$ Eigenvalues of $(\mathbf{A} - s\mathbf{I})^{-1}$ are $\nu_i = \frac{1}{\lambda_i - s}$ and $\lambda_{\max} = s + \nu_{\min}^{-1}$

Inverse Iteration Algorithm: Input $\mathbf{A}^{-1}$ in Power Iteration $\Rightarrow$ find $\lambda_{\min}^{-1}$ and $\mathbf{x}_{\min}$ (用 SII $s = 0$ 可直接得到)

Shifted Inverse Iteration Algorithm: Using $(\mathbf{A} - s\mathbf{I})^{-1}$ $\Rightarrow$ Find eigenvalue closest to s and corresponding eigenvector. Assume: $|\lambda_a - s| < |\lambda_b - s| \le \cdots$

Rate of Converge. for PI & SII: Rate $\uparrow$ iff $|\frac{\lambda_2}{\lambda_1}| \downarrow$ $\|\sigma_k\mathbf{z}^{(k)} - \mathbf{x}_1\|_2 \le \alpha_1(|\frac{\lambda_2}{\lambda_1}|)^k$; $|\lambda^{(k)} - \lambda_1| \le \alpha_2(|\frac{\lambda_2}{\lambda_1}|)^{2k}$ Rate $\uparrow$ iff $|\frac{\lambda_a - s}{\lambda_b - s}| \downarrow$ $\|\sigma_k\mathbf{z}^{(k)} - \mathbf{x}_a\|_2 \le \beta_1(|\frac{\lambda_a - s}{\lambda_b - s}|)^k$; $|\lambda^{(k)} - \lambda_a| \le \beta_2(|\frac{\lambda_a - s}{\lambda_b - s}|)^{2k}$

4.1 Orthogonal Iteration

For PI, II, SII, OI: Convergence of *eigenvalues* faster than *eigenvectors*.

Orthogonal Iteration: **Convergence:** If $|\lambda_1| > |\lambda_2| > \cdots > |\lambda_n|$ (distinct). **Speed of Conv.:** depend on $\max_{1\le l\le n-1} |\frac{\lambda_{l+1}}{\lambda_l}|$ It's **unstable**.

QR-Eig: **Convergence** and **Speed of Conv.** 和 OI 类似, 但实际上可以弱一点. **More stable** than OI. (more accurate)

Computational Cost(有优化过): a. **MM** Alg. for $\Lambda^{(k)}$, any **QR** Alg. $\to C = \mathcal{O}(k_{\max}n^3)$ b. **MV** Alg. for $\Lambda^{(k)}$, **MGS** for QR $\to C = \mathcal{O}(4n^3 + n^2 + n)$ c. **MM** Alg. $\to$ better alg. as $\mathbf{R}$ upper triangular.

DP Dot product $C=\sum_{i=1}^{n}2=2n=\mathcal{O}(n)$

Input: $\mathbf{x},\mathbf{y}\in\mathbb{R}^n$
Output: $s=\mathbf{x}\cdot\mathbf{y}=\mathbf{x}^\top\mathbf{y}$

$s=0$
for $i=1,\dots,n$ **do**
  $s=s+x_iy_i$
end for

MV Matrix-vector multiplication
$C=\sum_{i=1}^{m}2n=2nm=\mathcal{O}(mn)$

Input: $\mathbf{A}\in\mathbb{R}^{m\times n},\mathbf{x}\in\mathbb{R}^n$
Output: $\mathbf{y}=\mathbf{Ax}$

for $i=1,\dots,m$ **do**
  $y_i=\mathbf{a}_i^\top\mathbf{x}$ # **DP** Algorithm
end for

MM Matrix-matrix multiplication
$C=\sum_{i=1}^{m}\sum_{j=1}^{p}2n=2mnp=\mathcal{O}(mnp)$

Input: $\mathbf{A}\in\mathbb{R}^{m\times n},\mathbf{B}\in\mathbb{R}^{n\times p}$
Output: $\mathbf{C}=\mathbf{AB}$

for $i=1,\dots,m$ **do**
  for $j=1,\dots,p$ **do**
    $c_{ij}=\mathbf{a}_i^\top\mathbf{b}_j$ # **DP** Algorithm
  end for
end for

Rank Theorem: $[\mathbf{Ax}=\mathbf{0}$ iff $\mathbf{x}=\mathbf{0}]\Leftrightarrow$ [列线性无关] $\Leftrightarrow$ [$\mathbf{A}$ 可逆] $\Leftrightarrow$ [rank($\mathbf{A}$) = rank($\mathbf{A}^\top$) = n] **Abel:** ≥5 次方 | 实系数多项式 |∃实根无法用有限次加减乘除和开方表示.
Spectral Theorem: If $\mathbf{A}$ is symmetric, then ∃ orthonormal $\mathbf{Q}$ s.t. $\mathbf{A}=\mathbf{Q}\Lambda\mathbf{Q}^\top$ (Orthogonally diagonalisable) $\Lambda=\mathrm{diag}(\lambda_1,\dots,\lambda_n)$

LU LU factorisation
$C(n)=\sum_{k=1}^{n-1}(n-k)(1+2(n-k+1))=\frac{2}{3}n^3+\frac{1}{2}n^2-\frac{7}{6}n$

Input: $\mathbf{A}\in\mathbb{R}^{n\times n}$
Output: $\mathbf{L},\mathbf{U}\in\mathbb{R}^{n\times n}$ s.t. $\mathbf{A}=\mathbf{LU}$

$\mathbf{L}=\mathbf{I},\ \mathbf{U}=\mathbf{A}$
for $k=1,\dots,n-1$ **do**
  for $j=k+1,\dots,n$ **do**
    $l_{jk}=u_{jk}/u_{kk}$
    $(u_{jk},\dots,u_{jn})=(u_{jk},\dots,u_{jn})-l_{jk}(u_{kk},\dots,u_{kn})$
  end for
end for

FS Forward substitution
$C(n)=\sum_{j=1}^{n}[2(j-1)+1]=n^2=\mathcal{O}(n^2)$

Input: $\mathbf{L}\in\mathbb{R}^{n\times n}$ (lower triangular), $\mathbf{b}\in\mathbb{R}^n$
Output: $\mathbf{y}\in\mathbb{R}^n$ such that $\mathbf{Ly}=\mathbf{b}$

for $j=1,\dots,n$ **do**
  $y_j=\frac{1}{l_{jj}}\left(b_j-\sum_{k=1}^{j-1}l_{jk}y_k\right)$
end for

BS Back substitution
$C(n)=\sum_{j=1}^{n}[2(n-j)+1]=n^2=\mathcal{O}(n^2)$

Input: $\mathbf{U}\in\mathbb{R}^{n\times n}$ (upper triangular), $\mathbf{y}\in\mathbb{R}^n$
Output: $\mathbf{x}\in\mathbb{R}^n$ such that $\mathbf{Ux}=\mathbf{y}$

for $j=n,\dots,1$ **do**
  $x_j=\frac{1}{u_{jj}}\left(y_j-\sum_{k=j+1}^{n}u_{jk}x_k\right)$
end for

GE Gaussian elimination $C(n)=\frac{2}{3}n^3+\frac{5}{2}n^2-\frac{7}{6}n=\mathcal{O}(n^3)$

Input: $\mathbf{A}\in\mathbb{R}^{n\times n}$, $\mathbf{b}\in\mathbb{R}^n$
Output: $\mathbf{x}\in\mathbb{R}^n$ s.t. $\mathbf{Ax}=\mathbf{b}$

Find LU factorisation of $\mathbf{A}=\mathbf{LU}$. # LU Algorithm
Solve $\mathbf{Ly}=\mathbf{b}$ using forward substitution. # FS Algorithm
Solve $\mathbf{Ux}=\mathbf{y}$ using back substitution. # BS Algorithm

GEPP GE(partial pivoting) $C(n)=\frac{2}{3}n^3+3n^2-\frac{5}{3}n=\mathcal{O}(n^3)$

Input: $\mathbf{A}\in\mathbb{R}^{n\times n}$, $\mathbf{b}\in\mathbb{R}^n$ # numpy.linalg.solve use it
Output: $\mathbf{x}\in\mathbb{R}^n$ satisfying $\mathbf{Ax}=\mathbf{b}$

Find $\mathbf{PA}=\mathbf{LU}$ using Algorithm LUPP # LUPP Algorithm
Solve $\mathbf{L}\,\mathbf{y}=\mathbf{P}\,\mathbf{b}$ by forward substitution # FS Algorithm
Solve $\mathbf{U}\,\mathbf{x}=\mathbf{y}$ by back substitution # BS Algorithm

GECP GE(complete pivoting) $C(n)=n^3+3n^2-2n=\mathcal{O}(n^3)$

Input: $\mathbf{A}\in\mathbb{R}^{n\times n}$, $\mathbf{b}\in\mathbb{R}^n$
Output: $\mathbf{x}\in\mathbb{R}^n$ satisfying $\mathbf{A}\,\mathbf{x}=\mathbf{b}$

Find $\mathbf{P}\,\mathbf{A}\,\mathbf{Q}=\mathbf{L}\,\mathbf{U}$ using Algorithm LUCP # LUCP Algorithm
Solve $\mathbf{L}\,\mathbf{y}=\mathbf{P}\,\mathbf{b}$ by forward substitution # FS Algorithm
Solve $\mathbf{U}\,\mathbf{z}=\mathbf{y}$ by back substitution # BS Algorithm
Compute $\mathbf{x}=\mathbf{Q}\,\mathbf{z}$

LUPP LU factorisation with partial pivoting (not unique) $C(n)=\frac{2}{3}n^3+n^2-\frac{5}{3}n$

Input: $\mathbf{A}\in\mathbb{R}^{n\times n}$
Output: $\mathbf{L},\ \mathbf{U},\ \mathbf{P}\in\mathbb{R}^{n\times n}$ s.t. $\mathbf{PA}=\mathbf{LU}$ with $\mathbf{L}$ 单位下三角, $\mathbf{U}$ 可逆上三角

$\mathbf{U}=\mathbf{A},\ \mathbf{L}=\mathbf{I},\mathbf{P}=\mathbf{I}$
for $k=1,\dots,n-1$ **do**
  Choose $i\in\{k,\dots,n\}$ which maximizes $|u_{ik}|$ # **Max** Algorithm
  Exchange row $(u_{kk},\dots,u_{kn})$ with $(u_{ik},\dots,u_{in})$ # 第 k 行和第 i 行的 $k\sim n$ 列交换
  Exchange row $(l_{k1},\dots,l_{k,k-1})$ with $(l_{i1},\dots,l_{i,k-1})$ # 第 k 行和第 i 行的 $1\sim k-1$ 列交换
  Exchange row $(p_{k1},\dots,p_{kn})$ with $(p_{i1},\dots,p_{in})$ # 第 k 行和第 i 行的全部列交换
  for $j=k+1,\dots,n$ **do**
    $l_{jk}=u_{jk}/u_{kk}$
    $(u_{jk},\dots,u_{jn})=(u_{jk},\dots,u_{jn})-l_{jk}(u_{kk},\dots,u_{kn})$
  end for
end for

LUCP LU factorisation with complete pivoting (not unique) $C(n)=n^3+n^2-2n$

Input: $\mathbf{A}\in\mathbb{R}^{n\times n}$
Output: $\mathbf{L},\ \mathbf{U},\ \mathbf{P},\ \mathbf{Q}\in\mathbb{R}^{n\times n}$ s.t. $\mathbf{PAQ}=\mathbf{LU}$ with $\mathbf{L}$ 单位下三角, $\mathbf{U}$ 可逆上三角

$\mathbf{U}=\mathbf{A},\ \mathbf{L}=\mathbf{I},\ \mathbf{P}=\mathbf{I},\ \mathbf{Q}=\mathbf{I}$
for $k=1,\dots,n-1$ **do**
  Choose $(i,j)\in\{(k,k),\dots,(n,n)\}$ which maximizes $|u_{ij}|$ # **Max** Algorithm
  Exchange row $(u_{kk},\dots,u_{kn})$ with $(u_{ik},\dots,u_{in})$ # 第 k 行和第 i 行的 $k\sim n$ 列交换 U
  Exchange row $(l_{k1},\dots,l_{k,k-1})$ with $(l_{i1},\dots,l_{i,k-1})$ # 第 k 行和第 i 行的 $1\sim k-1$ 列交换 L
  Exchange row $(p_{k1},\dots,p_{kn})$ with $(p_{i1},\dots,p_{in})$ # 第 k 行和第 i 行的全部列交换 P
  Exchange column $(u_{1k},\dots,u_{nk})$ with $(u_{1j},\dots,u_{nj})$ # 第 k 列和第 j 列的全部行交换 U
  Exchange column $(q_{1k},\dots,q_{nk})$ with $(q_{1j},\dots,q_{nj})$ # 第 k 列和第 j 列的全部行交换 Q
  for $m=k+1,\dots,n$ **do**
    $l_{mk}=u_{mk}/u_{kk}$
    $(u_{mk},\dots,u_{mn})=(u_{mk},\dots,u_{mn})-l_{mk}(u_{kk},\dots,u_{kn})$
  end for
end for

(M)GS SQR Householder-QR

SQR (SLE) via QR-Factorization $C(n)=2n^3+4n^2+n$

Input: $\mathbf{A}\in\mathbb{R}^{n\times n},\mathbf{b}\in\mathbb{R}^n$
Output: $\mathbf{x}\in\mathbb{R}^n$ with $\mathbf{Ax}=\mathbf{b}$

Find QR factorization of $\mathbf{A}$ # **GS|MGS** Algorithm
Solve $\mathbf{Qy}=\mathbf{b}$ # **MV** Algorithm # ps: $\mathbf{y}=\mathbf{Q}^{-1}\mathbf{b}=\mathbf{Q}^\top\mathbf{b}$
Solve $\mathbf{Rx}=\mathbf{y}$ using Algorithm BS # **BS** Algorithm

Sum Notation: $\sum_{k=1}^{n}k=\frac{n(n+1)}{2}=\frac{1}{2}n^2+\frac{1}{2}n$
$\sum_{k=1}^{n}k^2=\frac{n(n+1)(2n+1)}{6}=\frac{1}{3}n^3+\frac{1}{2}n^2+\frac{1}{6}n$
$\sum_{k=1}^{n}k^3=\left(\frac{n(n+1)}{2}\right)^2=\frac{1}{4}n^4+\frac{1}{2}n^3+\frac{1}{4}n^2$
Rank Theorem: $\mathbf{A}_{m\times n}$, $m\geq n$: $[\mathbf{Ax}=\mathbf{0}$ iff $\mathbf{x}=\mathbf{0}]\Leftrightarrow[\mathrm{rank}(\mathbf{A})=n]$
Cauchy-Schwarz Inequality: For $\mathbf{x},\mathbf{y}\in\mathbb{R}^n$, $|\mathbf{x}\cdot\mathbf{y}|=|\mathbf{x}^\top\mathbf{y}|\leq\|\mathbf{x}\|_2\|\mathbf{y}\|_2$
Determinant: $\det(\mathbf{AB})=\det(\mathbf{A})\det(\mathbf{B})$ **Orthogonal:** If $\mathbf{Q}^\top\mathbf{Q}=\mathbf{I}$
Eigen: $\mathbf{Av}=\lambda\mathbf{v}$ $\xi_{\mathbf{A}}(\lambda)=\det(\mathbf{A}-\lambda\mathbf{I})=\prod_i(\lambda_i-\lambda)$

Iterative Method | OP MAX

GI|JAC|GAU JAC: $C=\mathcal{O}(k_{\max}n^2)$ | GAU: $C=\mathcal{O}(k_{\max}n^2)$

Input: $\mathbf{A}=\mathbf{M}+\mathbf{N}\in\mathbb{R}^{n\times n},\mathbf{b}\in\mathbb{R}^n,\mathbf{x}_0\in\mathbb{R}^n,\varepsilon_r>0$
Output: $\mathbf{x}_k\in\mathbb{R}^n$ with $\mathbf{Ax}_k\approx\mathbf{b}$

$k=0$ # JAC: $C=2n^2$/it GAU: $C=2n^2$/it
while $\|\mathbf{Ax}_k-\mathbf{b}\|_p>\varepsilon_r$ **do** # e.g. $\varepsilon_r=10^{-3}\|\mathbf{b}\|_p$
  $k=k+1$ # 上面给的 C 不含这步
  Compute $\mathbf{y}_k=\mathbf{b}-\mathbf{Nx}_{k-1}$
  Solve $\mathbf{Mx}_k=\mathbf{y}_k$ # ↓ $\mathbf{A}=\mathbf{D}+\mathbf{L}+\mathbf{U}$
end while # JAC: $M=D,N=L+U$; GAU: $M=D+L,N=U$

(M)GS (Modified) Gram-Schmidt $C(n)=2n^3+n^2+n$

Input: $\mathbf{Q}\in\mathbb{R}^{n\times n}$, with columns $\mathbf{a}_1,\dots,\mathbf{a}_n$
Output: $\mathbf{Q}\in\mathbb{R}^{n\times n}$, with orthonormal columns $\mathbf{q}_1,\dots,\mathbf{q}_n$
$\mathbf{R}\in\mathbb{R}^{n\times n}$ non-singular upper triangular matrix; s.t. $\mathbf{A}=\mathbf{QR}$

$\mathbf{R}=\mathbf{0}$
for $j=1,\dots,n$ **do**
  $\mathbf{q}_j=\mathbf{a}_j$
  for $k=1,\dots,j-1$ **do**
    $r_{kj}=\mathbf{q}_k^\top\mathbf{a}_j$ | $(r_{kj}=\mathbf{q}_k^\top\mathbf{q}_j)$
    $\mathbf{q}_j=\mathbf{q}_j-r_{kj}\mathbf{q}_k$
  end for
  $r_{jj}=\|\mathbf{q}_j\|_2=\sqrt{\mathbf{q}_j^\top\mathbf{q}_j}$
  $\mathbf{q}_j=\frac{\mathbf{q}_j}{r_{jj}}$
end for

QR Householder QR factorization $C(n)=\frac{8}{3}n^3+\mathcal{O}(n^2)$

Input: $\mathbf{A}\in\mathbb{R}^{n\times n}$
Output: $\mathbf{Q}\in\mathbb{R}^{n\times n}$ orthogonal
$\mathbf{R}\in\mathbb{R}^{n\times n}$ non-singular upper triangular matrix; s.t. $\mathbf{A}=\mathbf{QR}$

1: $\mathbf{Q}=\mathbf{I},\mathbf{R}=\mathbf{A}$ # 如果 $A\in\mathbb{R}^{m\times n}$, $m\geq n$, loop 到 n, 且 3.6.7 的 n 改为 m
2: **for** $k=1,\dots,n-1$ **do**
3:   $\mathbf{u}=(r_{kk},\dots,r_{nk})\in\mathbb{R}^{n-k+1}$
4:   $\mathbf{v}=\mathbf{u}+\mathrm{sign}(u_1)\|\mathbf{u}\|_2\mathbf{e}_1$ # line.4-7 得到的 $\mathbf{Q}_k$:
5:   $\mathbf{w}=\frac{\mathbf{v}}{\|\mathbf{v}\|_2}$ # 使得 $\mathbf{Q}_k\mathbf{u}=-\mathrm{sign}(u_1)\|\mathbf{u}\|_2\mathbf{e}_1$
6:   $\mathbf{H}_k=\mathbf{I}-2\mathbf{ww}^\top\in\mathbb{R}^{(n-k+1)\times(n-k+1)}$ # line.6-8 一起算, 见下
7:   $\mathbf{Q}_k=\mathbf{I}_{k-1}\oplus\mathbf{H}_k\in\mathbb{R}^{n\times n}$ # ↓line.6-8 $C=4(n-k+1)^2+(n-k+1)$
8:   $\mathbf{R}=\mathbf{Q}_k\mathbf{R}$ # 用矩阵分块, 只有右下角 $\mathbf{H}_k\mathbf{R}_{k:n,k:n}$ 要算 → 用 H_k 定义 Alg. MV+OP
9:   $\mathbf{Q}=\mathbf{QQ}_k$ # If using in Alg. **SQR**, then $9'$: $\mathbf{y}=\mathbf{Q}_k\mathbf{y}$; $C=\frac{4}{3}n^3+\mathcal{O}(n^2)$ (SQR/QR)
10: **end for** # $\mathbf{A}\in\mathbb{R}^{m\times n}$ reduced 的 $C=2n^2m-\frac{2}{3}n^3+\mathcal{O}(m^2+n^2+mn)$

OP Outer product $C=\sum_{i=1}^{n}\sum_{j=1}^{n}1=n^2=\mathcal{O}(n^2)$

Input: $\mathbf{x},\mathbf{y}\in\mathbb{R}^n$
Output: $A=\mathbf{xy}^\top$

for $i=1,\dots,n$ **do**
  for $j=1,\dots,n$ **do**
    $a_{ij}=x_iy_j$
  end for
end for

Max Finding index of maximum $C(m)=m-1$

Input: $z\in\mathbb{R}^m$
Output: index i with $|z_i|=\max_{j=1,\dots,m}|z_j|$

$i=1$, $\max=|z_1|$
for $j=2,\dots,m$ **do**
  if $|z_j|>\max$ **then**
    $i=j$
    $\max=|z_j|$
  end if
end for

LSQ-QR LSQ via QR $C=2mn^2-\frac{2}{3}n^3+\mathcal{O}(m^2+n^2+mn)$

Input: $\mathbf{A}\in\mathbb{R}^{m\times n}$, $m\geq n$, rank($\mathbf{A}$) $=n$, $\mathbf{b}\in\mathbb{R}^m$ ↑ by reduced QR
Output: $\mathbf{x}\in\mathbb{R}^n$ with $\|\mathbf{Ax}-\mathbf{b}\|_2$ minimised

Compute reduced QR factorization $\mathbf{A}=\mathbf{Q}_1\mathbf{R}_1$ # **QR** Algorithm
Compute $\mathbf{y}=\mathbf{Q}_1^\top\mathbf{b}$ # **MV** Algorithm # $C=2mn$
Solve $\mathbf{R}_1\mathbf{x}=\mathbf{y}$ # **BS** Algorithm # $C=n^2$

LSQ-SVD LSQ via SVD

Input: $\mathbf{A}\in\mathbb{R}^{m\times n}$, $m\geq n$, rank($\mathbf{A}$) $=n$, $\mathbf{b}\in\mathbb{R}^m$
Output: $\mathbf{x}\in\mathbb{R}^n$ with $\|\mathbf{Ax}-\mathbf{b}\|_2$ minimised

Compute reduced SVD $\mathbf{A}=\mathbf{U}_1\Sigma_1\mathbf{V}^\top$ # see SVD note
Compute $\mathbf{z}=\mathbf{U}_1^\top\mathbf{b}$ # **MV** Algorithm # $C=2mn$
Solve $\Sigma_1\mathbf{y}=\mathbf{z}$ # $C=n$
Compute $\mathbf{x}=\mathbf{Vy}$ # **MV** Algorithm # $C=2n^2$

QR-Eig QR Algorithm for Eigenvalues $C=$ 和 OI Alg. 类似

Input: $\mathbf{A}\in\mathbb{R}^{n\times n}$ *symmetric*, $\varepsilon>0$
Output: $\Lambda^{(k)}\in\mathbb{R}^{n\times n}$ with diagonal entries ≈ eigenvalues of $\mathbf{A}$.

$k=0$, $\Lambda^{(0)}=\mathbf{A}$
while $\|\Lambda^{(k)}-\Lambda^{(k-1)}\|_1>\varepsilon$ or $k=0$ **do**
  $k=k+1$ # By def $\mathbf{Q}^{(k)}=(\mathbf{Z}^{(k-1)})^\top\mathbf{Z}^{(k)}$, $\mathbf{Z}^{(k)}$ from **OI** Alg.
  Compute QR factorization $\Lambda^{(k-1)}=\mathbf{Q}^{(k)}\mathbf{R}^{(k)}$
  $\Lambda^{(k)}=\mathbf{R}^{(k)}\mathbf{Q}^{(k)}$
end while # Can solving $(\mathbf{A}-\lambda\mathbf{I})\mathbf{x}=\mathbf{0}$ to find eigenvector $\mathbf{x}$

PI Power Iteration $C=k_{\max}(4n^2+5n+1)$ (有优化过)

Input: $\mathbf{A}\in\mathbb{R}^{n\times n}$ *symmetric*, $\mathbf{z}^{(0)}\in\mathbb{R}^n\setminus\{\mathbf{0}\}$, $\varepsilon>0$
Output: $\mathbf{z}^{(k)}\in\mathbb{R}^n$, $\lambda^{(k)}\in\mathbb{R}$ s.t. $\mathbf{z}^{(k)}\approx\mathbf{x}_{\max}$, $\lambda^{(k)}\approx\lambda_{\max}$

$k=0$
$\lambda^{(0)}=r_{\mathbf{A}}(\mathbf{z}^{(0)})$
while $\|\mathbf{Az}^{(k)}-\lambda^{(k)}\mathbf{z}^{(k)}\|_2>\varepsilon$ **do** # or: $|\lambda^{(k)}-\lambda^{(k-1)}|>\varepsilon$
  $k=k+1$ # or: $\|\mathbf{z}^{(k)}-\mathbf{z}^{(k-1)}\|_2>\varepsilon$
  $\mathbf{z}^{(k)}=\mathbf{Az}^{(k-1)}$
  $\mathbf{z}^{(k)}=\frac{\mathbf{z}^{(k)}}{\|\mathbf{z}^{(k)}\|_2}$ # To improve stability and avoid large values
  $\lambda^{(k)}=r_{\mathbf{A}}(\mathbf{z}^{(k)})$ # In fact: $\mathbf{z}^{(k)}=\frac{\mathbf{A}^k\mathbf{z}^{(0)}}{\|\mathbf{A}^k\mathbf{z}^{(0)}\|_2}$ and $\lambda^{(k)}=r_{\mathbf{A}}(\mathbf{z}^{(k)})$
end while

SII Shifted Inverse Iteration $C=\mathcal{O}(n^3)+(k_{\max}-1)\mathcal{O}(n^2)$ (有优化过)

Input: $\mathbf{A}\in\mathbb{R}^{n\times n}$ *symmetric*, $s\in\mathbb{R}$, $\mathbf{z}^{(0)}\in\mathbb{R}^n\setminus\{\mathbf{0}\}$, $\varepsilon>0$
Output: $\mathbf{z}^{(k)}\in\mathbb{R}^n$, $\lambda^{(k)}\in\mathbb{R}$ s.t. $\mathbf{z}^{(k)}\approx\mathbf{x}_a$, $\lambda^{(k)}\approx\lambda_a$ near s

$k=0$
$\lambda^{(0)}=r_{\mathbf{A}}(\mathbf{z}^{(0)})$ # first $O(n^3)$ cost; then $O(n^2)$ cost/it
while $\|\mathbf{Az}^{(k)}-\lambda^{(k)}\mathbf{z}^{(k)}\|_2>\varepsilon$ **do**
  $k=k+1$
  Solve $(\mathbf{A}-s\mathbf{I})\mathbf{y}=\mathbf{z}^{(k-1)}$ for $\mathbf{y}$ # via GEPP & Find LU only once
  $\mathbf{z}^{(k)}=\frac{\mathbf{y}}{\|\mathbf{y}\|_2}$
  $\lambda^{(k)}=r_{\mathbf{A}}(\mathbf{z}^{(k)})$ # In fact: $\mathbf{z}^{(k)}=\frac{(\mathbf{A}-s\mathbf{I})^{-k}\mathbf{z}^{(0)}}{\|(\mathbf{A}-s\mathbf{I})^{-k}\mathbf{z}^{(0)}\|_2}$, $\lambda^{(k)}=r_{\mathbf{A}}(\mathbf{z}^{(k)})$
end while

OI Orthogonal Iteration $C=$ 取决于 QR 分解方法 |Λ 计算方法

Input: $\mathbf{A}\in\mathbb{R}^{n\times n}$ *symmetric*, $\mathbf{Z}^{(0)}\in\mathbb{R}^{n\times n}$ *orthogonal*, $\varepsilon>0$
Output: $\mathbf{Z}^{(k)}\in\mathbb{R}^{n\times n}$, $\Lambda^{(k)}\in\mathbb{R}^{n\times n}$ s.t.
$\mathbf{Z}^{(k)}\approx[\mathbf{x}_1,\dots,\mathbf{x}_n]$, $\Lambda^{(k)}\approx\mathrm{diag}(\lambda_1,\dots,\lambda_n)$

$k=0$
$\Lambda^{(0)}=(\mathbf{Z}^{(0)})^\top\mathbf{AZ}^{(0)}$
while $\|\mathbf{AZ}^{(k)}-\Lambda^{(k)}\mathbf{Z}^{(k)}\|_1>\varepsilon$ **do** # or: $\|\Lambda^{(k)}-\Lambda^{(k-1)}\|_1>\varepsilon$
  $k=k+1$
  Compute $\mathbf{W}^{(k)}=\mathbf{AZ}^{(k-1)}$
  Compute QR factorization $\mathbf{W}^{(k)}=\mathbf{Z}^{(k)}\mathbf{R}^{(k)}$
  $\Lambda^{(k)}=(\mathbf{Z}^{(k)})^\top\mathbf{AZ}^{(k)}$ # $\Lambda_{ii}^{(k)}=r_{\mathbf{A}}(\mathbf{z}_i^{(k)})$
end while # $\mathbf{Z}^{(k)}$ 第一列为 PI 结果

5 Appendix

5.1 Useful Theorems/Formulas

5.2 Others

Where the error come from calucation:

1. If your calculation involves 15 or more significant digits, you will likely encounter rounding errors. (Since $\varepsilon_m \approx 10^{-16}$ (relative accuracy), which is about 15 significant digits.)
2. **For GE Algorithm**: It is an unstable algorithm. e.g. $\mathbf{A} = [10^{-20}\ \ 1 \ \backslash\backslash \ \ 1\ \ 1]$ $\Rightarrow$ By **GE** Algorithm, $\widehat{\mathbf{L}}\widehat{\mathbf{U}} = \mathbf{A} + [0\ \ 0 \ \backslash\backslash \ \ 0\ \ -1]$ (error)
3. **MGS vs. GS**:In floating point arithmetic, the computed vectors $\{\hat{q}_1, \ldots, \hat{q}_{j-1}\}$ will not be orthonormal. Algorithm MGS takes this into account when computing $\hat{q}_j$; Algorithm GS does not.